

How Players Recognize and Interpret Diegetic Visualizations – The Case of Fortnite’s Swarm Visualization

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Abstract

Data visualization is a key part in many videogames as players need it to understand the current game state and react accordingly. While non-diegetic data visualizations in games have been widely studied, diegetic data visualizations have received little attention. In this work, we contribute to the latter by looking into the diegetic visualization used by *Fortnite* during Chapter 6: Season 4 to show how *The Swarm* spread over the map. Through an online survey with 80 participants, we explored how players recognized, interpreted, and acted upon this form of diegetic visualization. We found that participants were able to recognize and interpret the swarm areas through the diegetic indicators and that they adjusted their behavior based on them. However, the behavioral response varied based on the current in-game situation. We further reflect on some potential avenues for future research.

CCS Concepts

• **Applied computing** → **Computer games**; • **Human-centered computing** → *Information visualization*; *User studies*.

Keywords

Information Visualization, Fortnite, Diegetic Representations, User Study

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1 Introduction

Games need to convey various sorts of information to players through their interface to allow them to assess the current state

of the game and make decisions based on it [7, 21]. Towards this end, video games utilize various data visualization techniques to communicate information. These visualizations are, however, not standardized in games, as designers usually tailor them to their specific use cases [3, 22]. In addition, visualizations can be part of different elements of a game. Previous studies, such as Iacovides et al. [8] have distinguished between diegetic (relating to the narrative of the game, e.g., what the player character can see and interact with) and non-diegetic interface components (e.g., classic heads-up displays). Consequently, Hazzard [7] described diegetic data visualizations as something that exists in the game world, making sense to both the player and the player character, while non-diegetic data visualizations are part of the user interface which the player character can not see. Research has particularly focused on non-diegetic visualizations in games so far (e.g., [3, 22]). In case diegetic visualizations were of interest, discourse was usually centered around traditional visualization types integrated into the game world, for example, maps which are held by a player’s avatar (e.g., [5]). Indeed, some games might see an improvement in immersion and a player’s sense of control when non-diegetic visualizations are removed in favor of diegetic ones [8]. However, many games also use the game world itself to convey information, which can be considered a form of data visualization as well. These present an interesting use case as they are more subtle and might not be explicitly recognized as a data visualization in the first place.

An interesting recent example of such a form of visualization can be found in *Fortnite* [6]. The game has been studied for its connection to gaming disorders [9], effect on the social abilities of kids [1], utilization of telemetry data for game design [12], and players’ cardiac activity during gameplay [19]. We are, however, interested in how the game periodically alters the in-game world as part of its seasons. In doing so, the game constantly presents players with new visual information to discover and interact with [4, 13]. In particular, this paper reports an initial online study ($n = 80$) – employing rating and open-ended questions – to explore how players recognized, interpreted, and acted upon *Fortnite*’s visualization of how the *Swarm* spreads over the map (see Section 2), which indicated infested areas through visual elements such as blue smoke. In doing so, we aim to contribute to the understudied field of diegetic data visualization.



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Figure 1: Areas infested by the swarm have a distinct visual identity. (a, b) Aerial shots from the game. Blue smoke and gray hives serve as diegetic indicators of which areas are taken over by the insect creatures. (c) Close-up of a hive structure containing an orange bug-themed loot chest in the center.

2 Fortnite

Fortnite [6], is a third-person-shooter battle royale game. The game has players dropping in across a map, where they must look for weapons and supplies to become the last one standing to be victorious. Starting in August of 2025 with *Fortnite*'s Chapter 6: Season 4, the game introduced a new threat: insect-like creatures – the so-called *Onimorphs* – started spreading over the maps. As soon as an area has been taken over, it would be covered by blue smoke, gray hive structures, and orange nests containing loot, as shown in Figure 1. These visuals diegetically alerted the player about the *Onimorphs* as well as special loot (from the orange nests). Additionally, the game had quests that would require the player to recognize and enter one of these locations and complete tasks (such as eliminating *Onimorphs*, talking to non-player characters, etc.).

3 Methodology

Visual information first needs to be noticed, with saliency of visualizations being a much-discussed topic in the information visualization literature (e.g., [10]). Once the information is perceived, viewers need to be able to derive insight from it to, in turn, be able to act upon it [20]. Thus, we aim to explore the following research questions:

RQ1: Are players able to recognize the swarm areas?

RQ2: What information do they derive from the swarm areas?

RQ3: How did the players act in response to the swarm areas?

3.1 Survey

For the purposes of this study, we asked players who played *Fortnite* during Chapter 6: Season 4 (running from August 7th 2025 to November 1st 2025) to answer our survey. The survey was available immediately as the season ended on November 1, 2025, and was up until November 13. It took around 15 minutes to fill out and was advertised in *Fortnite* communities on Reddit (such as r/FortNiteLore and r/FortniteBattleRoyale) and several *Fortnite*-related Discord servers. In addition, we recruited participants via Prolific [16]. Participants on Prolific received an average compensation of £1.93 based on an hourly rate of £7.75, which was within Prolific's recommended payment range at the time of the study. The survey first gave a brief overview of the study and asked for informed consent. Participants had to be at least 18 years old to partake.

The survey itself consisted of three parts. The first part collected basic demographics (age, gender) and information related to their experience with the game (rank, number of play sessions, etc.). The second part contained various scale questions (see Table 1) related to the three research questions. These questions used a 7-point scale (1 = *Strongly Disagree* to 7 = *Strongly Agree*), asking participants to express their level of agreement. Finally, the survey included three non-mandatory open-ended questions, each one relating to one research question: *How did you recognize the swarm areas when you entered it?* (RQ1), *What information did you derive from the swarm areas on the map? How did you interpret them?* (RQ2), and *Did the swarm effect your gameplay decisions? If yes, how?* (RQ3).

3.2 Participants

A total of 114 responses were received. Of these, 34 responses were removed as participants indicated not having played the game recently (and thus not during the period of interest) or due to being incomplete. In addition, as the study was conducted online, we checked for unusually quick completion times using a z-score analysis with a threshold of 3 standard deviations [2]. This resulted in no exclusions, leading to a total sample of 80 participants for analysis. Qualitative analysis was based on the responses from 68 participants (as 12 did not provide qualitative feedback).

Participants' age ranged from 19 to 63 years with a mean age of 28 ($SD = 7.9$). In terms of gender, 47 participants self-identified as men, 32 as women, and 1 preferred not to disclose their gender. About 79% of our participants have reported to have reached at least *Gold* as the highest rank. *Gold* indicates a certain knowledge of the game and playtime, with *Gold* players being the 60th percentile this season [18].

3.3 Analysis

The text responses from the survey were analyzed using a content analysis process [11]. The process started with the first author inductively coding each of the three questions individually. Given the short nature of the responses, the codes were assigned on a participant level. This coding was then developed into an initial codebook, which was subsequently discussed between the first and last author. This revised codebook was then used to re-code the data. Based on this, the codebook was iterated one more time, and the coding was updated by the first author.

ID	Question
R1	I feel that the swarm areas were easily recognizable
R2	I noticed that the swarm areas changed over time across the map
I1	It was easy to understand which impact the swarm areas had on gameplay
I2	I felt the need to use external sources (like tutorials and wikis) to keep track of the swarm areas
A1	The swarm areas influenced my gameplay decisions
A2	I purposefully visited swarm areas
A3	I tried to stay away from swarm areas
A4	I kept track of the swarm areas to have an advantage in the game

Table 1: The rating questions used in the survey.

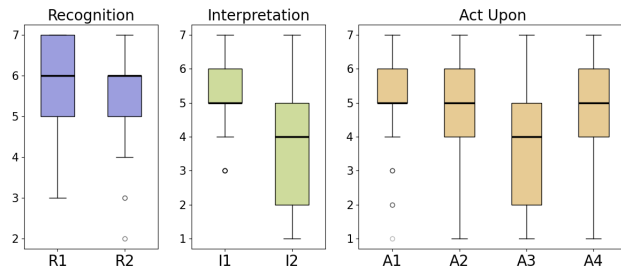


Figure 2: Ratings of the scale questions (see Table 1) organized along the three RQs.

4 Results

Figure 2 summarizes the ratings for the different questions listed in Table 1 while the results of the content analysis are shown in Table 2.

Recognition. Ratings showed that participants agreed that the areas were easily recognizable (R1) and that they noticed that they are not static but changed throughout the season (R2). The qualitative feedback revealed that they particularly (34 out of 68 responses, 50%) recognized them through their **DIEGETIC VISUAL INDICATORS** (gray bug hives, blue smoke, map destruction, etc.). However, nine participants (~13%) explicitly stated to have noticed the swarm areas in part or fully due to the **NON-DIEGETIC VISUAL INDICATORS** (minimap, pop-up text on screen, danger level meter, etc.). Further, about one-third of the participants (22, ~32%) mentioned that they also recognized the swarm areas in part or fully via the **AUDIO INDICATORS** (such as insect buzzing and intense music).

Interpretation. Quantitative responses indicated that the participants agreed that it was easy to understand the gameplay impact of the swarm areas (I1). There appears, however, to be less consensus about the need to use external sources (such as wikis and tutorials) to keep track of the swarm areas (I2). Qualitative feedback signaled that a third of the participants found the swarms to be useful information for improving their **POSITIONING** and movement around

the map (21, ~31%). Some participants were able to gauge a higher **ENEMY DENSITY** in the swarm areas (17, 25%), while others gathered a unique **LOOT TABLE** from them (9, ~13%). Some participants also interpreted these locations as **STORY** locations, as well as being the location for the weekly **QUESTS** (11, ~16%).

Act Upon. According to the ratings, participants somewhat agreed that the swarm areas influenced their in-game decisions (A1). Participants seemed to be somewhat in agreement about their decision to go to the swarm areas (A2), while they remain more split about avoiding swarm areas (A3). Participants also leaned towards keeping track of the swarm areas to secure an advantage in the game (A4). Qualitative responses indicated that a portion of the participants felt **ATTRACTED** towards the swarm areas (10, ~15%), for example, due to better loot drops. Others, in contrast, were **AVOIDANT** towards these areas as much as possible (11, ~16%) because of, for instance, the increased risk from the insect-like creatures and other players. The largest subset, however, expressed that they were both drawn to the swarm, most commonly by loot, but were also aware of the risk these areas offered, making their choice whether to approach them or not **SITUATIONAL**, depending on the various stages in the match, their own equipment, or perceived risk (30, ~44%). Some of the participants also reported that they did not change their gameplay because of the swarm areas (10, ~15%).

5 Discussion

In this paper, we focused on an initial exploration of *Fortnite*’s diegetic visualization of infested areas. In our view, these world changes, such as the addition of blue smoke, serve as a form of data representation, which could be viewed as a digital analogue to the concept of situated analytics [17] which refers to the use of data representation organized within the physical world. They serve as situated information source directly in the game which the players have to understand and make sense of to inform their gameplay decisions. However, there is currently little formal understanding on how these should be designed and how they shape behavior and experience.

Our participants reported, both through the ratings (R1, R2) and the qualitative feedback, that the swarm areas were easy to recognize (**RQ1**), particularly through their diegetic indicators – gray bug hives, blue smoke, and local map destruction – while the non-diegetic interface elements played a smaller role. The saliency of the visual in-game representation could be a factor here as it might render the additional interface elements less relevant. As mentioned earlier, saliency has been a major subject of interest in information visualization (e.g., [10]). As such it would be interesting to develop design guidelines on how such diegetic visualizations should be designed to be clearly observable without being distracting or breaking the coherence of the visual style, in other words, how to balance subtlety with legibility. Future work could also compare diegetic representations to other forms of cues to ascertain in which situations they are preferable or not.

Quantitative responses and I1 suggest that players generally found it easy to interpret the areas (**RQ2**). Participants commented that they interpreted them as being indicative of the *danger level* of the area, as they showed that the insect-like creatures concentrated there, but they also understood that these areas offer better loot

	Code	Description	#P
Recognition	DIEGETIC VISUAL INDICATORS	Statements about recognizing the swarm areas through their diegetic visual indicators	34
	NON-DIEGETIC VISUAL INDICATORS	Statements about recognizing the swarm areas through their non-diegetic visual indicators	9
	AUDIO INDICATORS	Statements about recognizing the swarm areas through audio cues	22
Interpretation	POSITIONING	Statements about participants changing their positioning based on the information they derived from the swarm areas	21
	ENEMY DENSITY	Statements about participants deriving enemy density information from the swarm areas	17
	LOOT TABLE	Statements about deriving information about unique loot from the swarm areas	9
	STORY AND QUESTS	Statements about deriving story elements and objectives of story quests from the swarm areas	11
Act Upon	NO CHANGE	Statements expressing no change to the gameplay	10
	ATTRACTED	Statements expressing an intention to go towards the swarm areas	10
	AVOIDANT	Statements expressing an intention to avoid the swarm areas	11
	SITUATIONAL	Statements about going to and avoiding the swarm areas depending on the game state	30

Table 2: Results of the content analysis of the open-ended feedback structured along the three RQs. #P represents the number of participants who reported a certain topic.

opportunities. In addition, players viewed them as valuable information for informing their positioning on the map. However, there was less consensus whether external sources are needed to keep track of them (I2). This could point to certain shortcomings of the diegetic indicators (see also Peacocke et al. [14]). It could also be indicative of a certain information-foraging [15] behavior where certain players try to find more information at extra cost through referring to external sources if the provided in-game signals are not considered sufficient. Indeed, information-foraging theory could be an interesting lens to study how players seek information in games, to then communicate the data at the right granularity.

Participants overall tended to agree that swarm areas influenced their decisions (A1) (RQ3) and that they tended to keep track of them to gain advantages (A4). This appears reflective of the idea introduced by Moore and Carter [13] where players were able to integrate their knowledge of the game world into their game skill expression. However, their behavioral responses varied: while some purposefully moved towards them (A2), for example, to get better loot, others tried to stay away from them (A3) to avoid the increased risk. However, the open-ended feedback showed that the majority of our participants made these decisions based on the stage of the match, their own equipment, or perceived risk. As such, the information conveyed by the diegetic representation does not take on a constant meaning, but rather a dynamic one that depends on the current situation. This situational interpretation could be an interesting avenue for future research, as the communicated data can take on a different meaning depending on the context.

Limitations. In this work, we only explored one diegetic representation, which, in terms of information conveyed, was mainly binary in the sense that it indicated whether a part of the map was infested or not. Future work should explore other and more complex

representations to develop a more comprehensive understanding of how such representations are perceived to, in turn, formulate design guidelines for such diegetic visualizations. This would also require assessing a more exhaustive range of information visualization properties (e.g., recency, readability) in more depth. With respect to the analysis, we need to acknowledge that the coding of the qualitative data was performed by a single researcher. However, to limit subjective bias, the codebook (together with a subset of the coding) was reviewed by a second researcher twice. Given the exploratory and very context-specific nature of the visualization, we also used a custom questionnaire. Our sample, judging from the ranks, also consisted of more experienced players and results may vary based on the knowledge of the game.

6 Conclusion

In this paper, we reported on a small study to explore how the swarm representation of *Fortnite* is perceived by players, by viewing it as a diegetic data visualization of the hostility/reward of an area. Such diegetic visualizations, i.e. visualizations that use elements of the game world instead of traditional charts, can be found in different games but have yet to receive more attention from the games and play community. While small in scope our study provides an initial exploration of one use case, showing that players can make good use of such visual indicators if designed properly. However, more research is needed to study such representations more systematically using a broader range of methods such as through the inclusion of telemetry analysis and observation to gain richer insights into how players engage with diegetic visualizations and how they are shaping their experience. This, in turn, will help to better understand their design space and to formulate actionable design guidelines.

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